HOG + SVM

Bottle crack patch classification report

Run: 20261003_154339

Labels: good = 0; crack = 1.

Existing dataset splits were retained. Exact duplicate image pixels across splits were checked. This does not verify physical-bottle separation.

Split	Good images	Crack images
train	315	70
valid	105	14
test	32	8

Model and selection procedure

Resize: (128, 128); grayscale input.

HOG settings: {'orientations': 9, 'pixels_per_cell': (8, 8), 'cells_per_block': (2, 2), 'block_norm': 'L2-Hys', 'transform_sqrt': True, 'feature_vector': True}

Features per image: 8100.

StandardScaler and SVM were fitted on train only. Kernel, C, class weight and decision threshold were selected using validation crack F1.

Selected settings: {'kernel': 'linear', 'C': 0.01, 'gamma': 'scale', 'class_weight': None}

Selected decision threshold: -0.006273

The SVM decision score is not a probability. Probability calibration and log loss are not included.

The selected model remains fitted on train only. Training scores are resubstitution scores; validation scores are selection scores; test scores are held-out.

Evaluation metrics

Metric	Train	Valid	Test
Accuracy	1.0000	0.9916	0.9500
Balanced accuracy	1.0000	0.9952	0.9219
Crack precision	1.0000	0.9333	0.8750
Crack recall	1.0000	1.0000	0.8750
Crack F1	1.0000	0.9655	0.8750
Specificity	1.0000	0.9905	0.9688
ROC AUC	1.0000	0.9993	0.9922
Average precision (AP)	1.0000	0.9952	0.9705

Split	TP	TN	FP	FN
train	70	315	0	0
valid	14	104	1	0
test	7	31	1	1

Average precision summarizes the precision-recall curve; it is not the trapezoidal PR area.

These results measure patch classification. Whole-bottle inspection requires a separate scanning and aggregation evaluation.

Runtime and model size

Total tuning time: 10.12 seconds.

Mean patch pipeline time: 19.70 ms over 40 test images.

Timing includes image reading, resizing, HOG extraction, scaling and SVM classification, with warm-up excluded. Drive access and hardware affect these measurements.

Model size: 8.60 MB.

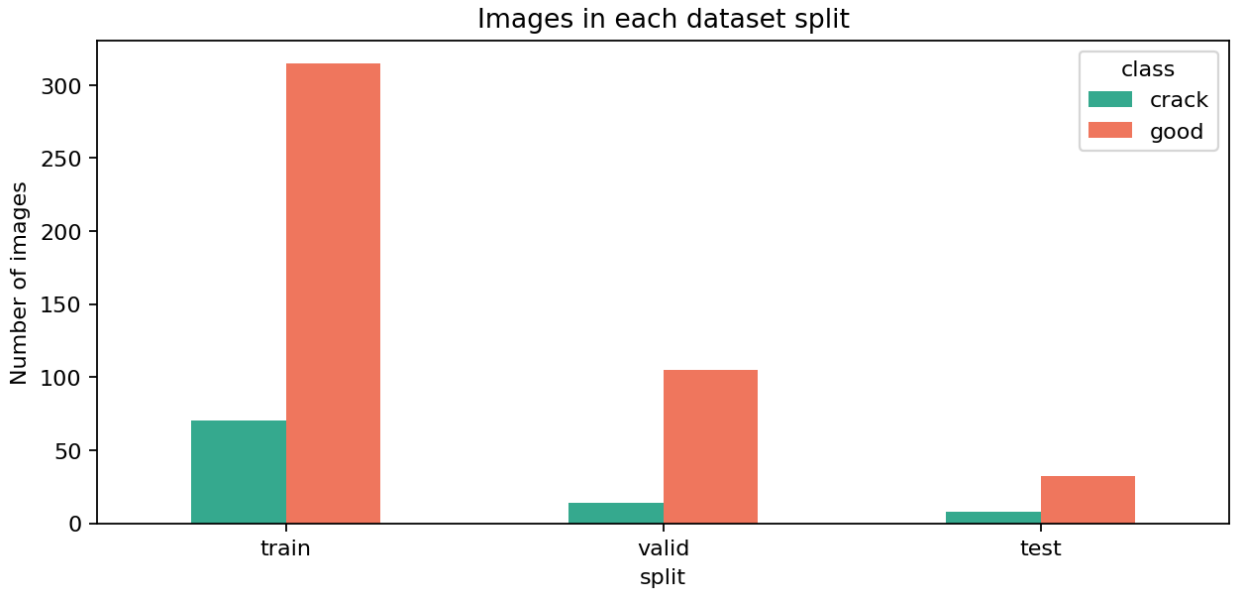
Support vectors: 136.

Python: 3.13.15; scikit-learn: 1.6.1; scikit-image: 0.25.2; Matplotlib: 3.10.0.

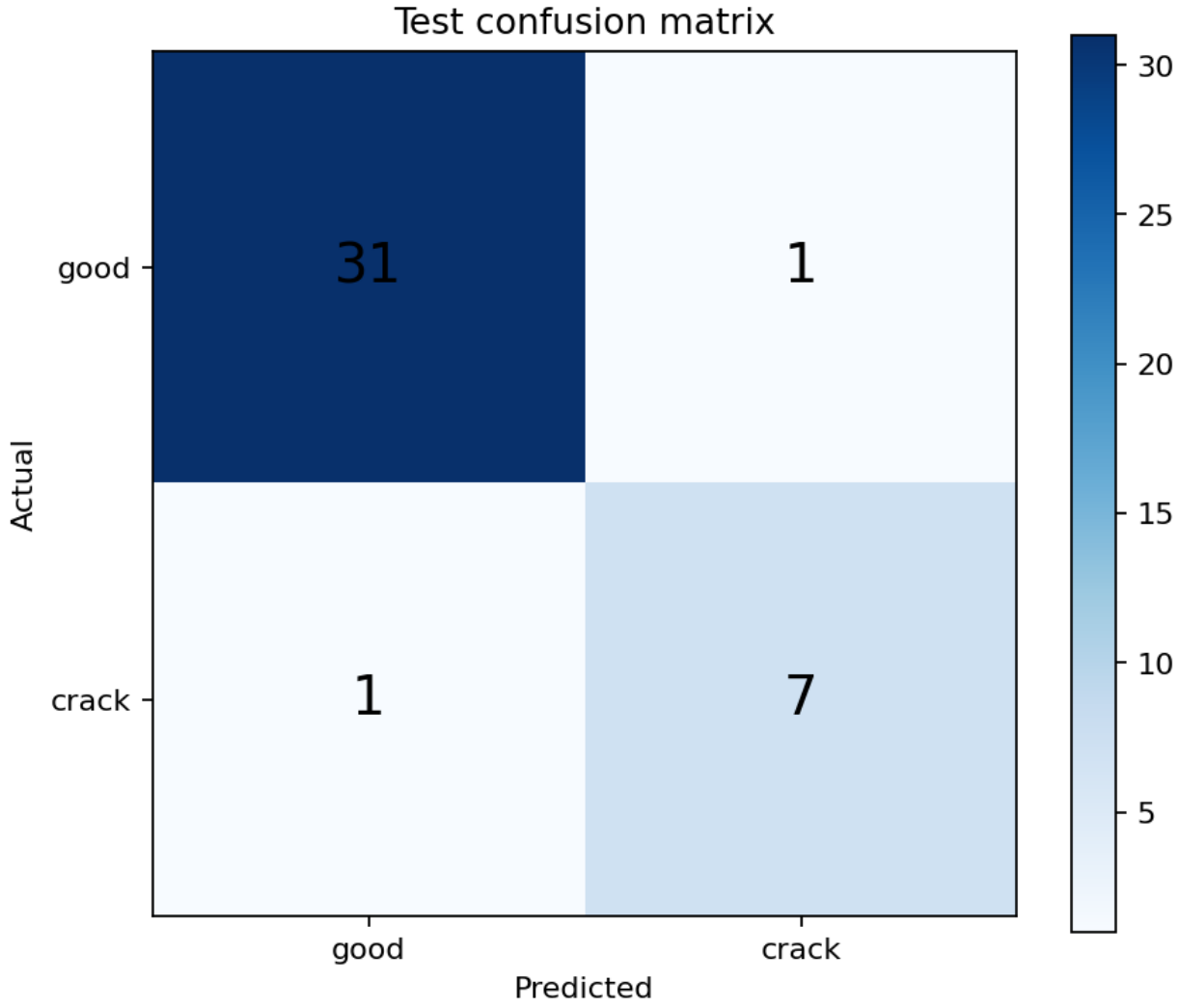
Validation search results

Kernel	C	Weight	Threshold	Valid F1
linear	0.01	None	-0.0063	0.9655
linear	0.01	balanced	-0.0063	0.9655
linear	0.1	None	-0.0063	0.9655
linear	0.1	balanced	-0.0063	0.9655
linear	1.0	None	-0.0063	0.9655
linear	1.0	balanced	-0.0063	0.9655
linear	10.0	None	-0.0063	0.9655
linear	10.0	balanced	-0.0063	0.9655
rbf	0.01	None	-0.9799	0.9032
rbf	0.01	balanced	-0.8297	0.8750
rbf	0.1	None	-0.7991	0.9032
rbf	0.1	balanced	0.1518	0.9032
rbf	1.0	None	0.0379	0.9333
rbf	1.0	balanced	0.0864	0.9333
rbf	10.0	None	0.0679	0.9333
rbf	10.0	balanced	0.0679	0.9333

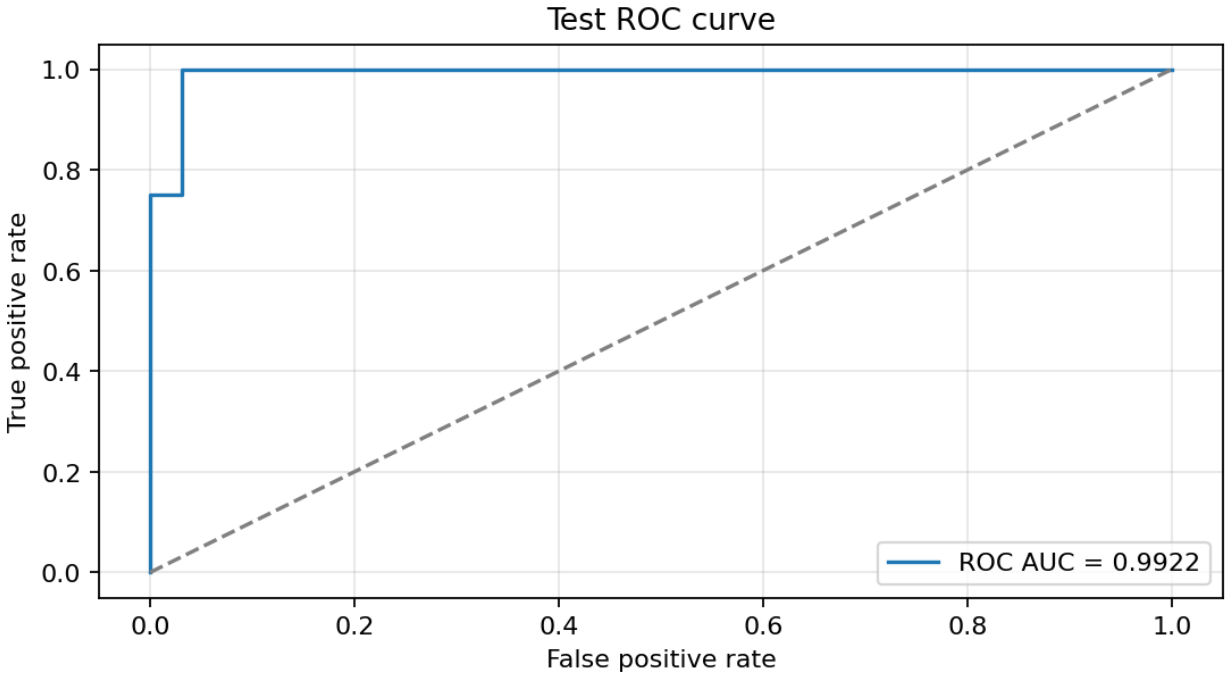
Dataset Distribution



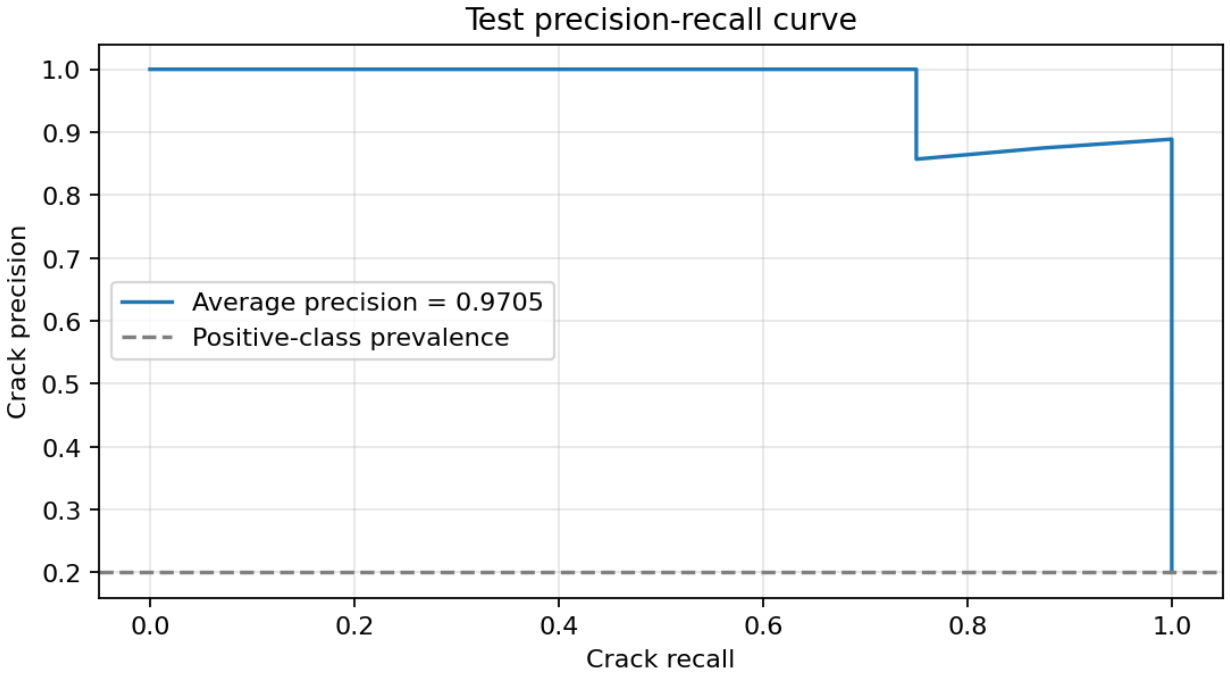
Test Confusion Matrix



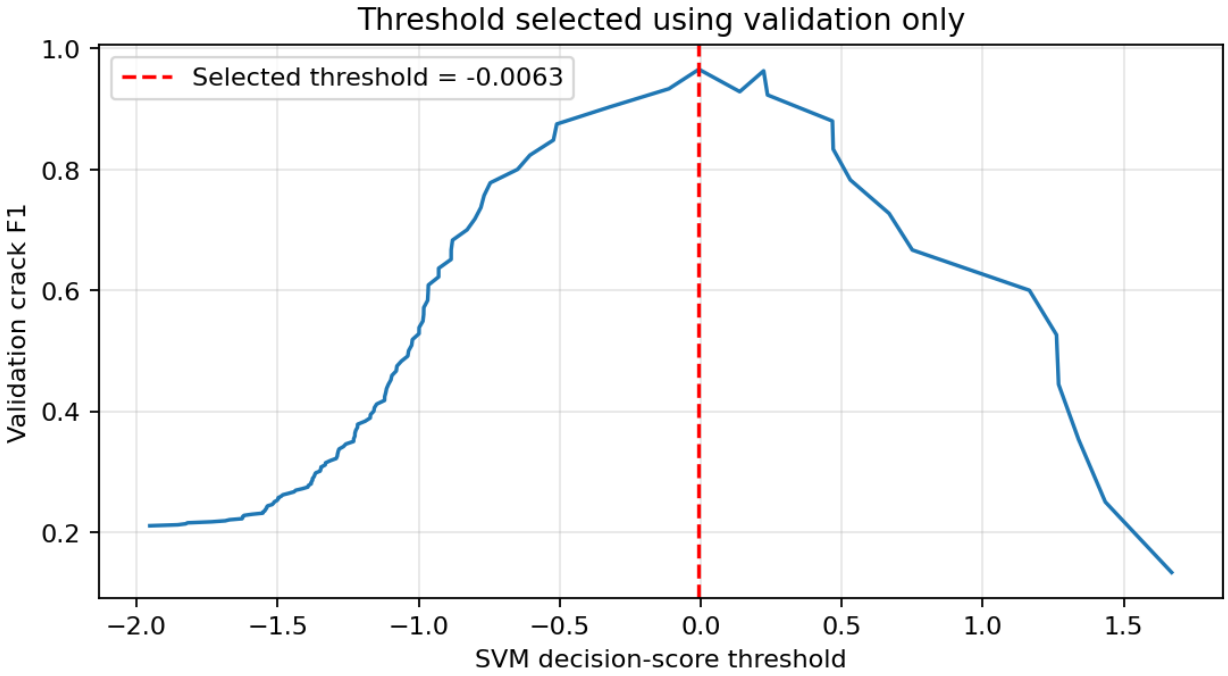
Test Roc


Test ROC curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC AUC = 0.9922

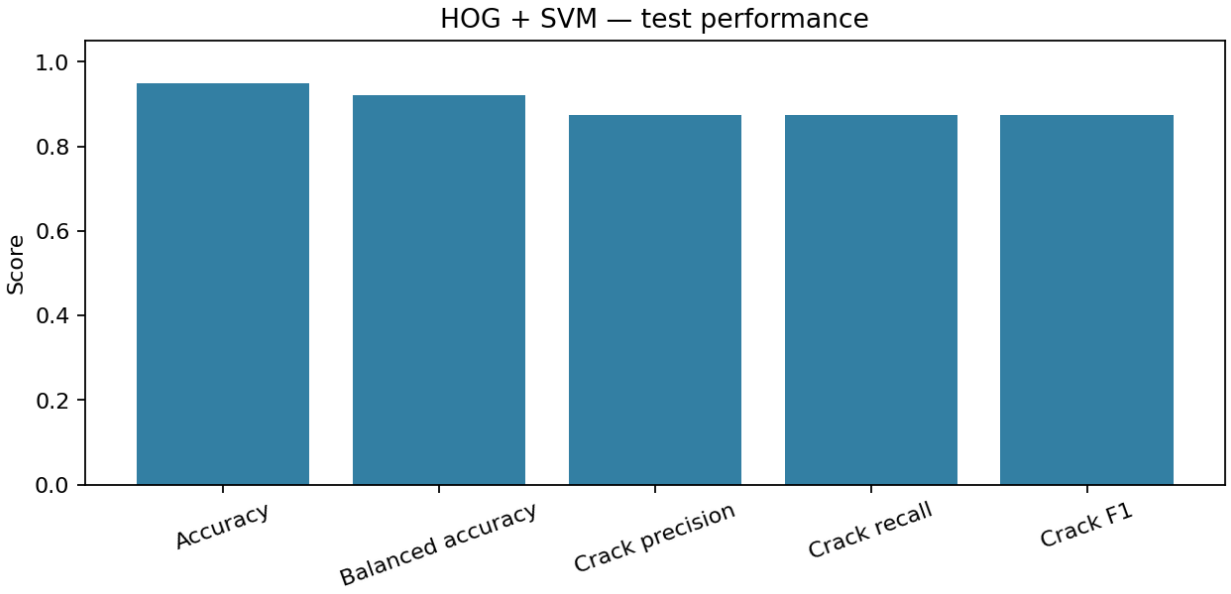
Test Precision Recall



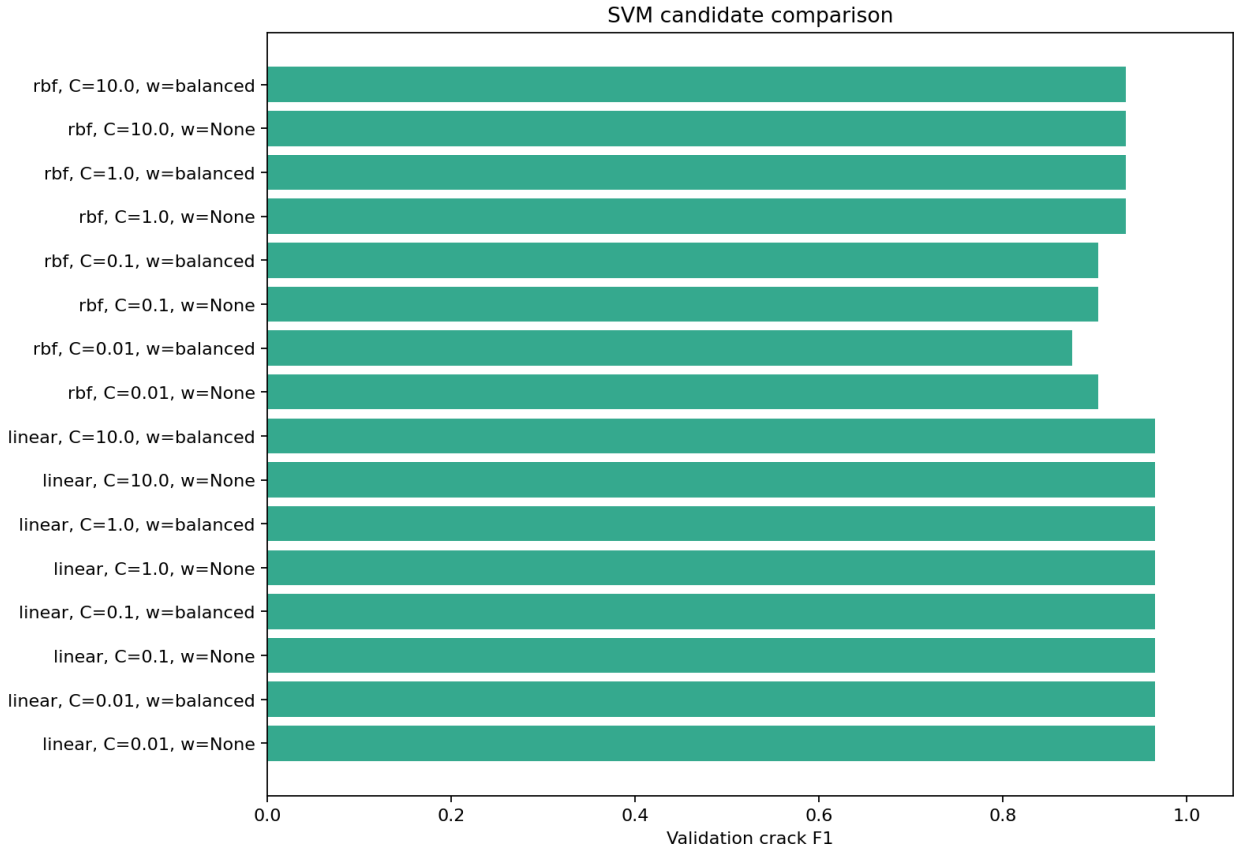
Validation Threshold



Test Metrics



Validation Search



Hog Example

Preprocessed training patch



HOG visualization

